



Subscription and Billing SaaS for Banking and Fintech

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Abstract

- Multi-tenant SaaS platform for banking and fintech companies to manage subscription billing and financial operations through one centralized system covering customers, plans, lending, billing, payments, support and reporting.
- Built with React, Node.js/Express, FastAPI, SQLAlchemy and MySQL to provide a structured, secure and scalable platform for financial service workflows.

Problem Statement & Objectives

- Problem:** Customer, plan, loan, repayment, billing, payment, refund and reporting workflows are often disconnected, making information hard to track and increasing operational complexity.
- Objectives**
 - Centralized platform for banking and fintech operations.
 - Multiple financial companies with separate administrators and customers.
 - Subscription plans, billing, invoices, payments and refunds.
 - Loan applications, loan management, repayment scheduling and tracking.
 - Company-level data isolation and role-based access.
 - Dashboards, search, notifications, documents, support and reporting.
- Scope**
 - Customers, company administrators, plans, subscriptions, loans, repayments, billing, payments, refunds, notifications, documents, support, search and reports.

Existing vs Proposed System

Feature	Existing Approach	Proposed System
Centralized Management	Operations may be handled through separate workflows.	Centralized platform for financial service operations.
Customer Management	Customer information may be spread across different processes.	Centralized customer management.
Loan Management	Loan and repayment tracking can need separate workflows.	Integrated loan application, loan, schedule and repayment management.
Billing & Payments	Billing, invoices and payments may be managed separately.	Integrated subscriptions, invoices, payments and refunds.
Multi-Tenancy	Separate systems or manual organization may be required.	Company-level tenant isolation with multiple administrators and customers.

Feasibility Study

- Technical:** Widely adopted web technologies: React, Node.js/Express, FastAPI and MySQL.
- Economic:** Open-source technologies and commonly available development tools.
- Operational:** Separate customer and company-admin workflows; operations centralized in one platform.
- Schedule:** Academic PBL project completed within the semester timeline.

Results & Testing

- 23 financial calculation tests passed.
- Backend Python modules compiled; application imports successfully.
- Database migrations generated and applied successfully.
- FastAPI application and OpenAPI routes load successfully.
- Loan, repayment, billing, refund, search, document, support and reporting routes implemented; API structure checked during development.

Conclusion

- Developed a centralized SaaS platform for banking and fintech financial operations.
- Integrated customer management, subscriptions, lending, billing, payments, refunds, support, documents, search and reporting.
- Implemented a multi-tenant architecture with structured backend services and MySQL-based data management.

Future Scope

- Real email and OTP delivery integration
- Google OAuth authentication
- External payment gateway integration
- Kafka-based event processing
- MongoDB / advanced document and search capabilities
- Spring Boot-based advanced reporting services
- Docker and Kubernetes deployment
- Prometheus and Grafana monitoring



GitHub

Tools & Technologies CURRENT IMPLEMENTATION

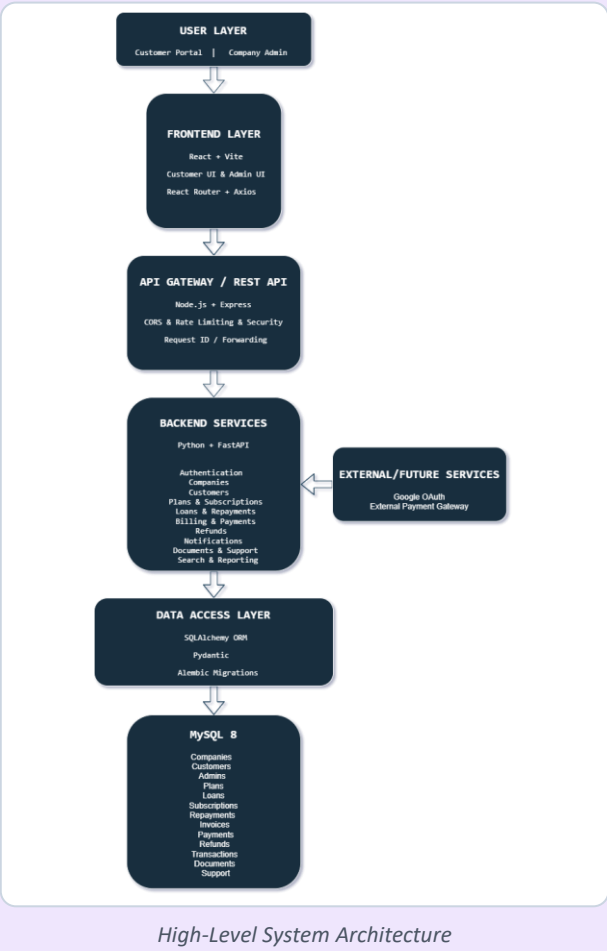
Frontend	React, Vite, Axios, React Router
API Gateway	Node.js, Express.js, Helmet, CORS, Express Rate Limit
Backend	Python, FastAPI, SQLAlchemy, Pydantic, Uvicorn
Database	MySQL 8
Dev / Testing Tools	Visual Studio Code, Git, GitHub, MySQL Workbench

Key Modules & Workflows

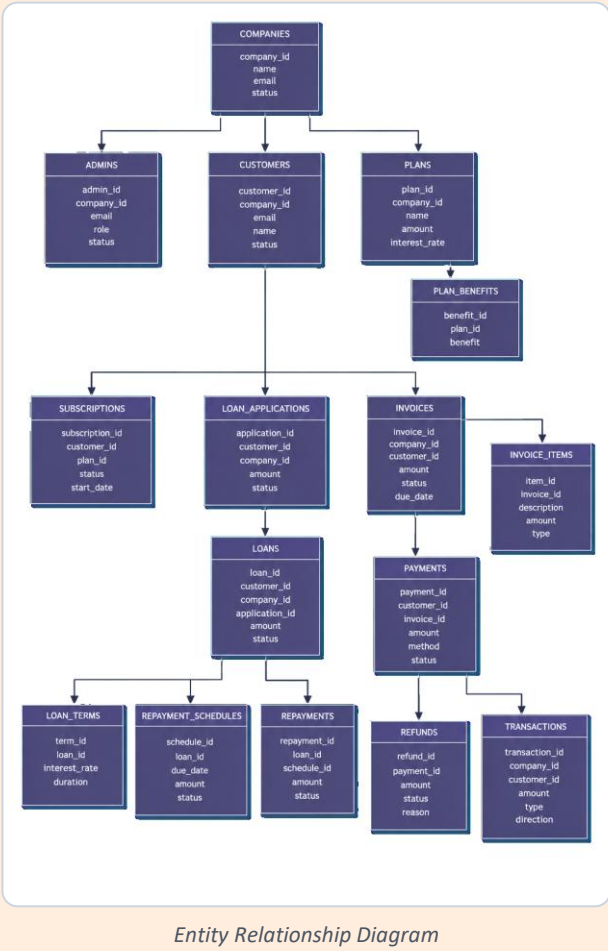
Authentication & Authorization	Company & Admin Management	Customer Management	Plans & Subscriptions
Loan Applications & Loans	Loan Repayments	Billing & Invoices	Payments & Refunds
Documents & Support Tickets	Search	Reporting & Exports	Notifications

Billing flow: Customer → Plan → Subscription → Invoice → Payment
Lending flow: Customer → Loan Application → Loan Approval → Loan → Repayment Schedule → Repayment

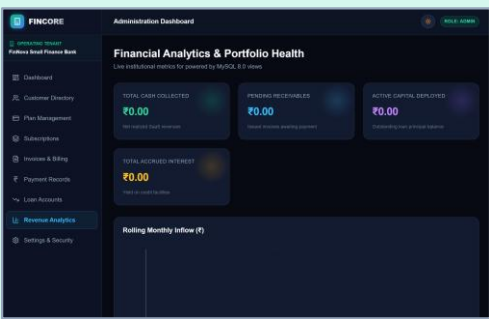
System Architecture



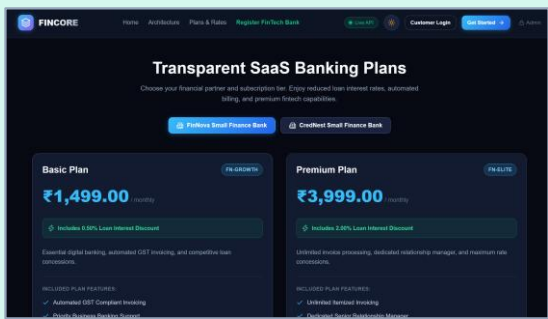
ER Diagram & Database Design



Implementation Screenshots



Administration Dashboard



Subscription Plans Page